

2. 设  $\boldsymbol{a}$  为常向量,  $\boldsymbol{r} = (x, y, z)$ ,  $r = |\boldsymbol{r}|$ ,  $f(r)$  为  $r$  的可微函数, 求下列各式之值:

(1)  $\nabla \cdot (r\boldsymbol{a})$ ;                      (2)  $\nabla \cdot (r^2\boldsymbol{a})$ ;

(3)  $\nabla \cdot (r^n\boldsymbol{a})$  ( $n$  为正整数);    ~~(4)~~  $\nabla \cdot \left(\frac{\boldsymbol{r}}{r^3}\right)$ ;

(5)  $\nabla \cdot [f(r)\boldsymbol{a}]$ ;                      (6)  $\nabla \cdot [\text{grad} f(r)]$ ;

~~(7)~~  $\nabla \cdot [f(r)\boldsymbol{r}]$ .

$$\begin{aligned} 2. (4) \quad \nabla \cdot \left(\frac{\vec{r}}{r^3}\right) &= \vec{r} \cdot \nabla \frac{1}{r^3} + \frac{1}{r^3} \nabla \cdot \vec{r} \\ &= \vec{r} \cdot \frac{-3}{r^4} \nabla \vec{r} + \frac{1}{r^3} (1+1+1) \\ &= \frac{-3\vec{r}}{r^4} \cdot \frac{\vec{r}}{r} + \frac{3}{r^3} \\ &= 0 \end{aligned}$$

$$\begin{aligned} (7) \quad \nabla [f(r) \cdot \vec{r}] &= f(r) \nabla \vec{r} + \vec{r} \nabla f(r) \\ &= 3f(r) + \vec{r} \cdot f'(r) \cdot \frac{\vec{r}}{r} \\ &= 3f(r) + r \cdot f'(r) \end{aligned}$$

5. 设数量场  $u(x, y, z)$  及向量场  $\boldsymbol{F}$  的各分量都有连续的二阶偏导数, 证明:

(1)  $\text{rot}(\text{grad} u) = \mathbf{0}$ ;                      (2)  $\text{div}(\text{rot} \boldsymbol{F}) = 0$ ;

~~(3)~~  $\text{rot}(u\boldsymbol{F}) = u \text{rot} \boldsymbol{F} + \nabla u \times \boldsymbol{F}$ .

6. 证明: 向量场  $\boldsymbol{F} = e^x (\cos y \sin z \boldsymbol{i} - \sin y \sin z \boldsymbol{j} + \cos y \cos z \boldsymbol{k})$  是保守场, 并求从点  $A(0, 0, \frac{\pi}{2})$  到  $B(1, \pi, \frac{\pi}{2})$  的曲线积分  $\int_{AB} \boldsymbol{F} \cdot d\boldsymbol{r}$ .

~~(7)~~ 证明下列向量场为有势场, 并求其势函数:

(1)  $\boldsymbol{F} = (-z \sin(xz) - y \cos(xy), -x \cos(xy), -x \sin(xz))$ ;

~~(2)~~  $\boldsymbol{F} = (yz \ln(1+z^2), xz \ln(1+z^2), xy \left[ \ln(1+z^2) + \frac{2z^2}{1+z^2} \right])$ .

6.  $p = e^x \cos y \sin z$ ,  $q = -e^x \sin y \sin z$ ,  $r = e^x \cos y \cos z$

$$\text{rot } \vec{F} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ p & q & r \end{vmatrix} = (0, 0, 0) = \vec{0}$$

则  $\vec{F}$  为保守场

$$f(x, y, z) = e^x \cos y \sin z \quad \int_{AB} \vec{F} \cdot d\vec{r} = e^x \cos y \sin z \Big|_A^B = -e^{-1}$$

$$5. (3) \text{rot}(u\vec{F}) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u p & u q & u r \end{vmatrix} = \left( \frac{\partial u}{\partial y} r + \frac{\partial r}{\partial y} u - \frac{\partial u}{\partial z} q - \frac{\partial q}{\partial z} u, \dots, \dots \right)$$

$$u \cdot \text{rot } \vec{F} + \nabla u \times \vec{F} = u \cdot \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ p & q & r \end{vmatrix} + \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ p & q & r \end{vmatrix} = \left( \frac{\partial r}{\partial y} u - u \frac{\partial q}{\partial z} + \frac{\partial u}{\partial y} r - \frac{\partial u}{\partial z} q, \dots, \dots \right)$$

∴

$$\therefore \text{rot}(u\vec{F}) = u \text{rot} \vec{F} + \nabla u \times \vec{F}$$



7. (2)  $\text{grad } \vec{F} = (-yz \ln(1+z^2), -xz \ln(1+z^2), -xy [\ln(1+z^2) + \frac{2z^2}{1+z^2}])$ , 故为有势场  
易看出  $f = -xyz \ln(1+z^2)$

9.1

$$\begin{cases} \frac{dx}{dt} = \cos wt \quad (w \neq 0) \\ x(0) = 10 \end{cases} \quad \begin{aligned} x(t) &= \frac{1}{w} \sin wt + C, \text{ 则 } C = 10 \\ x(t) &= \frac{1}{w} \sin wt + 10 \end{aligned}$$

9.2

(2)  $a \left( x \frac{dy}{dx} + 2y \right) = xy \frac{dy}{dx} \quad (a > 0);$   
 (3)  $\sqrt{1+x^2} dy - \sqrt{1-y^2} dx = 0;$   
 (4)  $(x+2y)dx + (2x-3y)dy = 0;$   
 (5)  $(3x+5y)dx + (4x+6y)dy = 0;$   
 (6)  $2x dz - 2z dx = \sqrt{x^2+4z^2} dx \quad (x > 0);$   
 (7)  $(2x^2+y^2)dx + (2xy+3y^2)dy = 0;$   
 (8)  $y' = (x+y+2)^2;$

(2)  $(ax-xy) \cdot \frac{dy}{dx} = -2ay$

①  $y \neq 0$

$$\frac{y-a}{2y} \cdot dy = \frac{a}{x} \cdot dx$$

$$\left(\frac{1}{2} - \frac{a}{2y}\right) dy = \frac{a}{x} dx$$

$$\frac{1}{2}(y - a \ln y) = a \ln x + C.$$

$$\frac{1}{2}y - C = \frac{1}{2}a \ln y + a \ln x$$

$$e^{\frac{1}{2}y+C} = y^{\frac{1}{2}a} \cdot x^a,$$

$$\therefore (x^2 y)^a = e^{y+C},$$

$$\text{即 } x^2 y = e^{\frac{y}{a}+C} = C e^{\frac{y}{a}}$$

②  $y \equiv 0$  时,

包含在  $C=0$  中

$$\text{故 } x^2 y = C e^{\frac{y}{a}}$$

(3)  $\sqrt{1+x^2} dy = \sqrt{1-y^2} dx$

①  $y \neq \pm 1$  时

$$\frac{1}{\sqrt{1-y^2}} dy = \frac{1}{\sqrt{1+x^2}} dx$$

$$\arcsin y = \ln(x + \sqrt{1+x^2}) + C.$$

②  $y = \pm 1$  时亦成立

则通解为  $\arcsin y = \ln(x + \sqrt{1+x^2}) + C$  或  $y = \pm 1$



$$(b). 2x dz - 2z dx = \sqrt{x^2 + 4z^2} dx$$

$$dz - \frac{2z}{2x} dx = \sqrt{\frac{1}{4} + \left(\frac{z}{x}\right)^2} dx$$

$$\text{令 } u = \frac{z}{x}, \quad z = ux, \quad \frac{dz}{dx} = u + x \frac{du}{dx}$$

$$u + x \frac{du}{dx} = u + \sqrt{u^2 + \frac{1}{4}}$$

$$\frac{1}{\sqrt{u^2 + \frac{1}{4}}} du = \frac{1}{x} dx$$

$$\ln(u + \sqrt{u^2 + \frac{1}{4}}) = \ln|x| + C, \quad u + \sqrt{u^2 + \frac{1}{4}} = C \cdot x$$

$$\frac{z}{x} + \sqrt{\left(\frac{z}{x}\right)^2 + \frac{1}{4}} = Cx,$$

$$\therefore z + \sqrt{z^2 + \frac{1}{4}x^2} = Cx^2, \text{ 即 } 2z + \sqrt{x^2 + 4z^2} = Cx^2$$

$$(8) \quad y' = (x + y + 2)^2.$$

$$\text{令 } z = x + y + 2, \quad \frac{dz}{dx} = 1 + \frac{dy}{dx} = 1 + z^2.$$

$$\therefore \frac{dz}{1+z^2} = dx$$

$$\arctan z = x + C.$$

$$\therefore \arctan(x + y + 2) = x + C$$

$$2. (3) \quad \sqrt{1+x^2} \frac{dy}{dx} = xy^3, \quad y(0) = 1$$

$$\textcircled{1} y \neq 0 \quad \frac{dy}{y^3} = \frac{x}{\sqrt{1+x^2}} dx$$

$$\frac{-1}{2y^2} = \frac{1}{2} \cdot 2 \cdot \sqrt{1+x^2} + C.$$

$$\text{即 } \frac{1}{y^2} = -2\sqrt{1+x^2} + C$$

$$-2y^2\sqrt{1+x^2} + Cy^2 = 1.$$

②  $y \equiv 0$ , 亦成立, 但不满足初值条件

$$y(0) = C(C-2) = 1, \quad C = 3$$

$$\therefore \text{原式为 } 3y^2 - 2y^2\sqrt{1+x^2} = 1$$



3. (2)

$$(2) \frac{dy}{dx} - \frac{2y}{x+1} = (x+1)^{5/2};$$

$$\frac{dy}{dx} - \frac{2y}{x+1} = (x+1)^{\frac{5}{2}}$$

$$P(x) = -\frac{2}{x+1}, Q(x) = (x+1)^{\frac{5}{2}}, \text{ 令 } a(x) = 0$$

$$\frac{dy}{dx} = \frac{2y}{x+1}, \quad \frac{dy}{2y} = \frac{dx}{x+1}, \quad \ln|y| = 2\ln|x+1| + C$$

$$y = (x+1)^2 \cdot C$$

$$\text{令 } C = u(x), \quad y = u(x) \cdot (x+1)^2, \quad \frac{dy}{dx} = u'(x) \cdot (x+1)^2 + u(x) \cdot 2 \cdot (x+1)$$

$$\text{代入原方程得 } u'(x) \cdot (x+1)^2 = (x+1)^{\frac{5}{2}}$$

$$u'(x) = (x+1)^{\frac{1}{2}}, \quad u(x) = \frac{2}{3}(x+1)^{\frac{3}{2}}$$

$$\therefore y = \frac{2C}{3}(x+1)^{\frac{7}{2}}.$$

4. 求下列微分方程的通积分：

$$(1) y' + \frac{y}{x} = y^3;$$

$$(2) xy' + y = 2\sqrt{xy};$$

$$(3) nxy' + 2y = xy^{n+1};$$

$$(4) 3xy' - y - 3xy^4 \ln x = 0;$$

$$(5) y' - 3xy - xy^2 = 0.$$

5. 将下列微分方程化为线性微分方程：

$$(1) \frac{dy}{dx} = \frac{x^2 + y^2}{2y};$$

$$(2) \frac{dy}{dx} = \frac{y}{x + y^2};$$

$$(3) 3xy^2 \frac{dy}{dx} + y^3 + x^3 = 0;$$

$$(4) \frac{dy}{dx} = \frac{1}{\cos y} + x \tan y.$$

$$4. (4) \quad 3xy' - y - 3xy^4 \ln x = 0$$

$$3x \frac{dy}{dx} \cdot \frac{1}{y^4} = \frac{1}{y^3} + 3x \ln x$$

$$-x \frac{dy^{-3}}{dx} = y^{-3} + 3x \ln x, \text{ 令 } z = y^{-3}$$

$$-x \cdot \frac{dz}{dx} = z + 3x \ln x$$

$$\therefore \frac{dz}{dx} + \frac{z}{x} = -3 \ln x, P(x) = \frac{z}{x}, Q(x) = -3 \ln x$$

$$\text{令 } Q(x) = 0, \quad \frac{dz}{dx} = -\frac{z}{x}, \quad \frac{dz}{z} = -\frac{dx}{x}, \quad \ln|z| = -\ln|x| + C, \quad z = C \cdot \frac{1}{x}$$

$$\text{令 } C = u(x), \quad z = u(x) \cdot \frac{1}{x}, \quad z' = u'(x) \cdot \frac{1}{x} - \frac{1}{x^2} \cdot u(x)$$

$$u'(x) \cdot \frac{1}{x} = -3 \ln x, \quad u'(x) = -3x \ln x$$



$$\begin{aligned}\int x \ln x \, dx &= \frac{1}{2} \int \ln x \, dx^2 = \frac{1}{2} (x^2 \ln x - \int x^2 d \ln x) \\ &= \frac{1}{2} (x^2 \ln x - \frac{1}{2} x^2) \\ &= \frac{1}{2} x^2 \ln x - \frac{1}{4} x^2\end{aligned}$$

$$\therefore u(x) = \frac{3}{4} x^2 - \frac{3}{2} x^2 \ln x + C$$

$$\therefore z = \frac{3}{4} x - \frac{3}{2} x \ln x + \frac{C}{x}$$

$$\therefore xy^{-3} - \frac{3}{4} x^2 + \frac{3}{2} x^2 \ln x = C$$

5.(4)

5. 将下列微分方程化为线性微分方程:

$$(1) \frac{dy}{dx} = \frac{x^2 + y^2}{2y};$$

$$(2) \frac{dy}{dx} = \frac{y}{x + y^2};$$

$$(3) 3xy^2 \frac{dy}{dx} + y^3 + x^3 = 0;$$

$$(4) \frac{dy}{dx} = \frac{1}{\cos y} + x \tan y.$$

$$\frac{dy}{dx} = \frac{1}{\cos y} + x \cdot \frac{\sin y}{\cos y}, \quad \frac{d \sin y}{dx} = 1 + x \sin y$$

令  $z = \sin y$ , 则

$$\frac{dz}{dx} = 1 + xz$$

6. 一曲线在点  $(x, y)$  处的斜率等于  $\frac{2y+x+1}{x}$ , 且通过点  $(1, 0)$ , 试求此曲线的表达式.

设  $l: y = f(x)$

$$\begin{cases} y' = \frac{2y+x+1}{x} \\ f(1) = 0 \end{cases}, \quad \Delta = \begin{vmatrix} 1 & 2 \\ 1 & 0 \end{vmatrix} = -2 \neq 0$$

$$\text{方程 } \begin{cases} 2y+x+1=0 \\ x=0 \end{cases} \text{ 有唯一解 } \begin{cases} x=0 \\ y=-\frac{1}{2} \end{cases}$$

$$\therefore y' = \frac{2(y+\frac{1}{2})+x}{x}, \quad \text{令 } y+\frac{1}{2} = y_1, \text{ 则有}$$

$$y_1' = \frac{2y_1+x}{x} = \frac{2y_1}{x} + 1, \quad \frac{dy_1}{dx} = \frac{2y_1}{x} + 1.$$



$$\text{令 } u = \frac{y_1}{x}, \quad y_1 = ux, \quad \frac{dy_1}{dx} = \frac{du}{dx} \cdot x + u.$$

$$\therefore \frac{du}{dx} \cdot x + u = 2u + 1,$$

$$\frac{du}{dx} \cdot x = u + 1$$

$$\frac{du}{u+1} = \frac{1}{x} dx, \quad \ln|u+1| = \ln|x| + C$$

$$u+1 = C \cdot x, \quad u = Cx - 1, \quad y_1 = Cx^2 - x,$$

$$y = Cx^2 - x - \frac{1}{2},$$

$$C = \frac{3}{2}$$

$$\therefore \text{b: } y = \frac{3}{2}x^2 - x - \frac{1}{2}$$

### 习 题 9.2

1. 求下列微分方程的通积分:

(1)  $\frac{dy}{dx} = \frac{1+y^2}{(1+x^2)xy}$ ;

(2)  $a \left( x \frac{dy}{dx} + 2y \right) = xy \frac{dy}{dx} \quad (a > 0)$ ;

(3)  $\sqrt{1+x^2} dy - \sqrt{1-y^2} dx = 0$ ;

(4)  $(x+2y)dx + (2x-3y)dy = 0$ ;

(5)  $(3x+5y)dx + (4x+6y)dy = 0$ ;

(6)  $2x dz - 2z dx = \sqrt{x^2+4z^2} dx \quad (x > 0)$ ;

(7)  $(2x^2+y^2)dx + (2xy+3y^2)dy = 0$ ;

(8)  $y' = (x+y+2)^2$ ;

(9)  $(2x+3y-1)dx + (4x+6y-5)dy = 0$ ;

(10)  $(2x-y+4)dy + (x-2y+5)dx = 0$ .

2. 求下列初值问题的解:

$$(1). \frac{y}{1+y^2} dy = \frac{dx}{x(1+x^2)}$$

$$\frac{1}{2} \ln|1+y^2| = \ln \left| \sin \frac{x}{\sqrt{1+x^2}} \right| + C$$

$$(1+y^2)^{\frac{1}{2}} = C \frac{x}{\sqrt{1+x^2}}$$

$$1+y^2 = C \cdot \frac{x^2}{1+x^2}$$

$$(1+y^2)(1+x^2) = Cx^2$$

$$(4). (x+2y)dx + (2x-3y)dy = 0$$

$$(1+2\frac{y}{x})dx + (2-\frac{3y}{x})dy = 0, \quad (1+2u)dx = (3u-2)dy, \quad \frac{dy}{dx} = \frac{1+2u}{3u-2} = \frac{du}{dx} \cdot x + u$$

$$\frac{y}{x} = u, \quad y = ux, \quad y' = u'x + u$$

$$\frac{1+2u-3u^2+2u}{3u-2} = \frac{du}{dx} \cdot x$$

$$\left( \frac{1+2u}{3u-2} - u \right) = \frac{du}{dx} \cdot x$$

$$\frac{1}{x(1+x^2)} dx = \frac{\cos \theta}{\tan \theta} \frac{1}{\cos^2 \theta} d\theta = \frac{1}{\tan \theta} d\theta = \frac{\cos \theta}{\sin \theta} d\theta = \frac{1}{\sin \theta} d\sin \theta = \ln |\sin \theta|$$

$$x = \tan \theta, \quad \theta = \arctan x,$$

$$\begin{array}{c} x \\ \diagup \\ \text{---} 1 \text{---} \end{array} \begin{array}{c} \sqrt{1+x^2} \\ \diagdown \\ \theta \end{array}, \quad \sin \theta = \frac{x}{\sqrt{1+x^2}}$$



$$\frac{dx}{x} = \frac{3u-2}{4u+1-3u^2} du$$

$$3u-2=v, \quad u = \frac{v+2}{3},$$

$$du = \frac{1}{3} dv.$$

$$= \frac{v}{-\frac{v^2+4v+4}{3} + \frac{4v+8}{3} + 1}$$

$$= \frac{v}{\frac{4-v^2}{3} + 1} \cdot \frac{1}{3} dv = \frac{v}{7-v^2} dv \quad \frac{1}{2} \frac{1}{7-v^2} dv^2$$

$$- \frac{1}{2} \ln |7-v^2|$$

$$- \frac{1}{2} (7 - (9u^2 - 12u + 4))$$

$$\ln x = - \frac{1}{2} \cdot \ln (12u - 9u^2 + 3) + C$$

$$x = C \cdot (12u - 9u^2 + 3)^{-\frac{1}{2}}$$

$$x^2 = C \cdot \frac{1}{12u - 9u^2 + 3}$$

$$\frac{1}{x^2} = C \cdot (4u - 3u^2 + 1)$$

$$\frac{1}{x^2} = C \cdot (4 \frac{y}{x} - 3 \frac{y^2}{x^2} + 1)$$

$$1 = C \cdot (4xy - 3y^2 + x^2)$$

$$4xy - 3y^2 + x^2 = C$$



1. 求下列微分方程的通积分:

$$(1) \frac{dy}{dx} = \frac{1+y^2}{(1+x^2)xy};$$

$$(2) a \left( x \frac{dy}{dx} + 2y \right) = xy \frac{dy}{dx} \quad (a > 0);$$

$$(3) \sqrt{1+x^2} dy - \sqrt{1-y^2} dx = 0;$$

$$(4) (x+2y)dx + (2x-3y)dy = 0;$$

$$(5) (3x+5y)dx + (4x+6y)dy = 0;$$

$$(6) 2x dz - 2z dx = \sqrt{x^2+4z^2} dx \quad (x > 0);$$

$$(7) (2x^2+y^2)dx + (2xy+3y^2)dy = 0;$$

$$(8) y' = (x+y+2)^2;$$

$$(9) (2x+3y-1)dx + (4x+6y-5)dy = 0;$$

$$(10) (2x-y+4)dy + (x-2y+5)dx = 0.$$

2. 求下列初值问题的解:

$$(7) -\frac{2x^2+y^2}{2xy+3y^2} = \frac{dy}{dx}$$

$$-\frac{2+u^2}{2u+3u^2} = x \cdot \frac{du}{dx} + u$$

$$-\frac{2+3u^2+3u^3}{2u+3u^2} = x \cdot \frac{du}{dx}$$

$$-\frac{1}{x} dx = \frac{2u+3u^2}{3u^3+3u^2+2} du$$

全微分计算

$$(9) -\frac{2x+3y-1}{4x+6y-5} = \frac{dy}{dx}$$

$$2x+3y=u, \quad 3y=u-2x, \quad 3\frac{dy}{dx} = \frac{du}{dx} - 2$$

$$-\frac{u-1}{2u-5} = \frac{1}{3} \frac{du}{dx} - \frac{2}{3}$$

$$-\frac{3u-3}{2u-5} = \frac{du}{dx} - 2.$$

$$\frac{4u-10-3u+3}{2u-5} = \frac{du}{dx}$$

$$\frac{u-7}{2u-5} = \frac{du}{dx}$$

$$dx = \frac{2u-5}{u-7} du$$

$$(15) (3x+5y) \cdot dx + (4x+6y) dy = 0$$

$$-\frac{3x+5y}{4x+6y} = \frac{dy}{dx}$$

$$\frac{y}{x} = u, \quad y = ux, \quad \frac{dy}{dx} = u'x + u.$$

$$-\frac{3+5u}{4+6u} = x \cdot \frac{du}{dx} + u$$

$$-\frac{3+9u+6u^2}{4+6u} = x \cdot \frac{du}{dx}$$

$$-\frac{dx}{x} = \frac{4+6u}{3+9u+6u^2} du$$

$$= \frac{2}{3} \frac{2+3u}{1+3u+2u^2} du$$

$$\frac{2+3u}{(2u+1)(u+1)}$$

$$= \frac{2}{3} \left( \frac{1}{2u+1} + \frac{1}{u+1} \right) du$$

$$-\ln x = \frac{1}{3} \ln(2u+1) + \frac{2}{3} \ln(u+1)$$

$$\frac{1}{x} = (1/3) \cdot (2u+1)^{\frac{1}{3}} \cdot (u+1)^{\frac{2}{3}}.$$

$$\frac{1}{x^3} = C \cdot (2u+1)(u+1)^2$$

$$1 = C \cdot (2y+x) \cdot (x+y)^2$$

$$(x+y)^2 \cdot (x+2y) = C$$



$$dx = \left(2 + \frac{9}{u-7}\right) du$$

$$x = 2u + 9 \ln(u-7) + C$$

$$x = 4x + 6y + 9 \ln(2x+3y-7)$$

$$x + 2y + 3 \ln(2x+3y-7) = C$$

(10)

$$(9) (2x+3y-1)dx + (4x+6y-5)dy = 0;$$

$$(10) (2x-y+4)dy + (x-2y+5)dx = 0.$$

$$(10) \frac{dy}{dx} = - \frac{x-2y+5}{2x-y+4} \quad \begin{vmatrix} 1 & -2 \\ 2 & -1 \end{vmatrix} = 3 \quad \begin{vmatrix} 1 & 5 \\ 2 & 4 \end{vmatrix} = -6$$

$$\begin{vmatrix} -2 & 5 \\ -1 & 4 \end{vmatrix} = -3$$

$$y = 2, \quad x = -1$$

$$\frac{dy}{dx} = - \frac{(x+1) - 2(y-2)}{2(x+1) - (y-2)}, \quad \begin{cases} x+1 = x_1 \\ y-2 = y_1 \end{cases}$$

$$\frac{dy_1}{dx_1} = - \frac{x_1 - 2y_1}{2x_1 - y_1} = - \frac{1 - 2\frac{y_1}{x_1}}{2 - \frac{y_1}{x_1}}, \quad \frac{y_1}{x_1} = u, \quad y_1 = ux_1, \quad y_1' = u'x_1 + u_1$$

$$- \frac{1-2u}{2-u} = \frac{du}{dx_1} \cdot x_1 + u$$

$$\frac{2u-1-2u+u^2}{2-u} = \frac{du}{dx_1} \cdot x_1$$

$$\frac{u^2-1}{2-u} = \frac{du}{dx_1} \cdot x_1,$$

$$\frac{dx_1}{x_1} = \frac{2-u}{u^2-1} du$$

$$= \frac{1}{2} \left( \frac{1}{u+1} - 3 \cdot \frac{1}{u-1} \right) du$$

$$\int \frac{2-u}{u^2-1} du,$$

$$\frac{u+1}{u+1} - \frac{u+1}{u-1} = \frac{2}{u^2-1}$$

$$\frac{1}{u+1} + \frac{1}{u-1} = \frac{2u}{u^2-1}$$

$$\int \frac{2}{u^2-1} du = \frac{1}{2} \cdot 2 \cdot \ln \left| \frac{u-1}{u+1} \right| = \ln \left| \frac{u-1}{u+1} \right| - \frac{1}{2} \ln |u^2-1|$$

$$\frac{1}{2} \frac{1}{u^2-1} du^2 = \frac{1}{2} \ln |u^2-1|$$



$$\ln x_1 = \ln \left| \frac{u-1}{(u+1)\sqrt{u^2-1}} \right| + C, \quad x_1 = C \left( \frac{u-1}{(u+1)\sqrt{u^2-1}} \right)$$

$$x_1 = C \cdot \frac{\frac{y_1}{x_1} - 1}{\left( \frac{y_1}{x_1} + 1 \right) \sqrt{\frac{y_1^2}{x_1^2} - 1}}, \quad x_1 = C \cdot \frac{y_1 - x_1}{(y_1 + x_1) \sqrt{y_1^2 - x_1^2}} = C_1 \cdot \frac{y_1 - x_1}{(y_1 + x_1) \sqrt{y_1 - x_1} \sqrt{y_1 + x_1}}$$

分母乘了两个  $x_1$ , 而不是一个! 所以  $x_1$  会除过去

$$\ln x_1^2 = \ln |u+1| - 3 \ln |u-1|$$

$$x_1 = C_1 \frac{\sqrt{y_1 - x_1}}{(y_1 + x_1)^{\frac{3}{2}}}$$

$$C \cdot x_1^2 \cdot (u-1)^3 = u+1$$

$$x+1 = C_1 \frac{\sqrt{y-x-3}}{(x+y-1)^{\frac{3}{2}}}$$

$$C \cdot x_1^2 \left( \frac{y_1}{x_1} - 1 \right)^3 = \frac{y_1}{x_1} + 1$$

$$(x+1) (x+y-1)^{\frac{3}{2}} = C_1 \sqrt{y-x-3}$$

$$C \cdot (y_1 - x_1)^3 = y_1 + x_1$$

$$(1) \frac{dx}{y} + \frac{4dy}{x} = 0, y(4) = 2;$$

$$(2) x dx + y e^{-x} dy = 0, y(0) = 1;$$

$$(3) \sqrt{1+x^2} \frac{dy}{dx} = x y^3, y(0) = 1;$$

$$(4) x \frac{dy}{dx} + 2y = \sin x, y(\pi) = \frac{1}{\pi}.$$

3. 求下列微分方程的通解:

$$(1) x \frac{dy}{dx} - y = (x-1)e^x;$$

$$(2) \frac{dy}{dx} - \frac{2y}{x+1} = (x+1)^{5/2};$$

$$(3) (x+1) \frac{dy}{dx} - n y = e^x (x+1)^{n+1};$$

$$(4) \frac{dy}{dx} + 2y = x e^{-x}.$$

$$(1) \frac{dx}{y} = -\frac{4dy}{x}, \quad x dx = -4y dy, \quad \frac{1}{2} x^2 = -2y^2 + C$$

$$\frac{1}{2} \times 16 = -2 \times 4 + C.$$

$$C = 16.$$

$$\therefore x^2 + 4y^2 - 36 = 0$$

$$(2) x dx = -y e^{-x} dy, \quad x e^x dx = -y dy,$$

$$e^x (x-1) = -\frac{1}{2} y^2 + C.$$

$$-1 = -\frac{1}{2} + C, \quad C = -\frac{1}{2}$$

$$\therefore y^2 + 2(x-1)e^x + 1 = 0$$

$$(4) \frac{dy}{dx} + 2 \frac{y}{x} = \frac{\sin x}{x}, \quad \frac{dy}{dx} + 2 \frac{y}{x} = 0$$



$$\frac{dy}{dx} = -\frac{2y}{x}, \quad \frac{dy}{y} = -\frac{2}{x} \cdot dx$$

$$\ln|y| = -2\ln|x| + C.$$

$$y = \frac{C}{x^2}, \quad \text{令 } C = u(x)$$

$$y = \frac{u(x)}{x^2}, \quad y'(x) = \frac{u'(x) \cdot x^2 - u(x) \cdot 2x}{x^4} = \frac{u'(x)x - 2u(x)}{x^3}$$

$$\frac{u'(x) \cdot x - 2u(x)}{x^3} + 2 \cdot \frac{u(x)}{x^3} = \frac{\sin x}{x}$$

$$\frac{u'(x)}{x^2} = \frac{\sin x}{x}$$

$$\frac{du}{dx} = x \sin x$$

$$du = x \sin x \, dx$$

$$-x \, d(\cos x) = -x \cos x + \cos x \, d(-x)$$

$$u = -x \cos x - \sin x + C$$

$$= -x \cos x - \sin x$$

$$y = \frac{C}{x^2} \cdot (-x \cos x - \sin x)$$

$$\frac{1}{2} = \frac{C}{2^2} \cdot (2) = \frac{C}{2}, \quad C = 1$$

$$\therefore y = \frac{-x \cos x - \sin x}{x^2}.$$

$$(1) \frac{dx}{y} + \frac{4dy}{x} = 0, \quad y(4) = 2;$$

$$(2) x \, dx + y e^{-x} \, dy = 0, \quad y(0) = 1;$$

$$(3) \sqrt{1+x^2} \frac{dy}{dx} = x y^3, \quad y(0) = 1;$$

$$(4) x \frac{dy}{dx} + 2y = \sin x, \quad y(\pi) = \frac{1}{\pi}.$$

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$$(3) (x+1) \frac{dy}{dx} - ny = e^x (x+1)^{n+1};$$

$$(4) \frac{dy}{dx} + 2y = x e^{-x}.$$

$$3. (1) \frac{dy}{dx} - \frac{1}{x} y = \frac{x-1}{x} e^x$$

$$\frac{dy}{dx} = \frac{y}{x}, \quad \frac{dy}{y} = \frac{dx}{x}, \quad \ln|y| = \ln|x| + C, \quad y = Cx$$

$$\text{令 } C = u(x), \quad y = u(x) \cdot x, \quad y' = u'(x)x + u(x)$$



$$\therefore u'(x) \cdot x = \frac{x-1}{x} e^x$$

$$u'(x) = \frac{x-1}{x^2} e^x, \quad u(x) = \frac{e^x}{x}$$

$$\therefore y = e^x + Cx$$

$$(3) \quad \frac{dy}{dx} - \frac{n}{x+1} y = e^x (x+1)^n \quad \frac{dy}{y} = \frac{n}{x+1} dx, \quad \ln|y| = n \cdot \ln|x+1|$$

$$\therefore y = C \cdot (x+1)^n, \quad \text{令 } C = u(x), \quad y = u(x)(x+1)^n,$$

$$y' = u'(x) \cdot (x+1)^n + u(x) \cdot n(x+1)^{n-1}$$

$$\therefore u'(x) \cdot (x+1)^n = e^x (x+1)^n, \quad u(x) = e^x + C.$$

$$\therefore y = (e^x + C) \cdot (x+1)^n.$$

$$(4) \quad \frac{dy}{dx} + 2y = xe^{-x}, \quad \frac{dy}{y} = -2y, \quad \frac{dy}{y} = -2 \cdot dx \quad \ln y = -2x + C, \quad y = C \cdot e^{-2x}$$

$$C = u(x), \quad y = u(x) \cdot e^{-2x}, \quad y'(x) = u'(x)e^{-2x} + u(x) \cdot -2 \cdot e^{-2x}$$

$$\therefore u'(x) \cdot e^{-2x} = xe^{-x}, \quad u'(x) = xe^x,$$

$$u(x) = e^x(x-1) + C,$$

$$\therefore y = e^{-x}(x-1) + Ce^{-2x}$$

4. 求下列微分方程的通积分:

$$(1) \quad y' + \frac{y}{x} = y^3;$$

$$(2) \quad xy' + y = 2\sqrt{xy};$$

$$(3) \quad nxy' + 2y = xy^{n+1};$$

$$(4) \quad 3xy' - y - 3xy^4 \ln x = 0;$$

$$(5) \quad y' - 3xy - xy^2 = 0.$$

5. 将下列微分方程化为线性微分方程:

$$(1) \quad \frac{dy}{dx} = \frac{x^2 + y^2}{2y};$$

$$(2) \quad \frac{dy}{dx} = \frac{y}{x + y^2};$$

$$(3) \quad 3xy^2 \frac{dy}{dx} + y^3 + x^3 = 0;$$

$$(4) \quad \frac{dy}{dx} = \frac{1}{\cos y} + x \tan y.$$

$$(2) \quad xy' + y = \sqrt{xy},$$

$$\frac{1}{2\sqrt{y}} \frac{dy}{dx} + \frac{y}{2\sqrt{y} \cdot x} = \frac{\sqrt{x}}{x}$$

$$\frac{d\sqrt{y}}{dx} + \frac{\sqrt{y}}{2x} = 2x^{-\frac{1}{2}}, \quad \sqrt{y} = z$$

$$\frac{dz}{dx} = -\frac{z}{2x}, \quad \frac{dz}{z} = -\frac{dx}{2x}, \quad \ln|z| = -\frac{1}{2}\ln|x| + C$$

$$\therefore z = C \cdot x^{-\frac{1}{2}}, \quad C = u(x)$$

$$4.(1) \quad \frac{1}{y^3} \cdot \frac{dy}{dx} + \frac{1}{xy^2} = 1$$

$$-\frac{1}{2} \frac{dy^{-2}}{dx} + \frac{1}{x} y^{-2} = 1, \quad y^{-2} = z,$$

$$-\frac{1}{2} \frac{dz}{dx} + \frac{z}{x} = 1$$

$$\frac{1}{2} \frac{dz}{dx} = \frac{z}{x} - 1, \quad \text{令 } \frac{z}{x} = u, \quad z = ux, \quad \frac{dz}{dx} = u'x + u.$$

$$\therefore \frac{du}{dx} \cdot x + u = 2u - 1, \quad \frac{du}{dx} = \frac{u-1}{x}$$

$$\therefore u-1 = Cx, \quad u = Cx + 1 = \frac{z}{x}.$$

$$z = Cx^2 + 2x = y^{-2}$$

$$Cx^2y^2 + 2xy^2 = 1$$



$$\frac{dz}{dx} = u'(x) x^{-\frac{1}{2}} + u(x) \cdot -\frac{1}{2} x^{-\frac{3}{2}},$$

$$u'(x) x^{-\frac{1}{2}} = x^{-\frac{1}{2}}, \quad u'(x)=1, \quad u(x)=C+x$$

$$z = (C+x) \cdot x^{-\frac{1}{2}}$$

$$y = (x+C)^2 \cdot x^{-1}, \quad xy = (x+C)^2$$

4. 求下列微分方程的通积分：

$$(1) \ y' + \frac{y}{x} = y^3;$$

$$(2) \ xy' + y = 2\sqrt{xy};$$

$$(3) \ nxy' + 2y = xy^{n+1};$$

$$(4) \ 3xy' - y - 3xy^4 \ln x = 0;$$

$$(5) \ y' - 3xy - xy^2 = 0.$$

5. 将下列微分方程化为线性微分方程：

$$(1) \ \frac{dy}{dx} = \frac{x^2 + y^2}{2y};$$

$$(2) \ \frac{dy}{dx} = \frac{y}{x + y^2};$$

$$(3) \ 3xy^2 \frac{dy}{dx} + y^3 + x^3 = 0;$$

$$(4) \ \frac{dy}{dx} = \frac{1}{\cos y} + x \tan y.$$

$$(5) \ \frac{1}{y^2} \frac{dy}{dx} - \frac{3x}{y} = x$$

$$- \frac{dy^{-1}}{dx} - 3xy^{-1} = x$$

$$\frac{1}{2} y^{-1} = z,$$

$$\frac{dz}{dx} + 3xz + x = 0$$

$$\frac{dz}{dx} = -x(3z+1)$$

$$\frac{dz}{3z+1} = -x dx$$

$$\ln|3z+1| = -\frac{3}{2}x^2 + C$$

$$\therefore 3z+1 = C \cdot e^{-\frac{3}{2}x^2}.$$

$$3y^{-1} = Ce^{-\frac{3}{2}x^2} - 1$$

$$Ce^{-\frac{3}{2}x^2} y - y = 3$$

$$(3) \ \frac{n \cdot x \cdot \frac{dy}{dx}}{y^{n+1}} + \frac{2}{y^n} = x$$

$$-x \cdot \frac{dy^{-n}}{dx} + 2 \cdot y^{-n} = x, \quad z = y^{-n},$$

$$\therefore -\frac{dz}{dx} + \frac{2z}{x} = 1$$

$$\frac{z}{x} = u, \quad \frac{dz}{dx} = u'x + u$$

$$\therefore \frac{du}{dx} \cdot x + u - 2u = -1 \quad \frac{du}{dx} = -\frac{u-1}{x}, \quad u-1 = Cx, \quad u = Cx+1$$

$$\therefore z = Cx^2 + x = y^{-n},$$

$$\therefore y^n(Cx^2 + x) = 1$$

$$5.(1) \quad \frac{dy^2}{dx} = x^2 + y^2, \quad \frac{dz}{dx} = x^2 + z$$

$$(2) \quad d(x+y^2) = dx + 2y dy$$

$$\frac{dx}{dy} = \frac{x+y^2}{y}$$

9. 求解积分方程

$$\int_0^1 \varphi(tx) dt = n\varphi(x),$$

其中  $\varphi(x)$  为可微函数.(提示：对左端积分做变量替换  $u=xt$ , 再两边求导得关于  $\varphi(x)$  的微分方程.)

10. 设  $y_1(x)$  与  $y_0(x)$  分别是线性非齐次方程(9.16)与其对应的齐次方程(9.17)的解, 证明:  $y_1(x) + y_0(x)$  也是非齐次方程(9.16)的解.

11. 设  $y_1(x), y_2(x)$  是非齐次方程(9.16)的两个解, 证明:  $y_1(x) - y_2(x)$  是(9.16)对应的齐次方程(9.17)的解.

$$9. \quad u=xt, \quad t=\frac{u}{x}, \quad dt=\frac{1}{x} du$$

$$\int_0^x \varphi(u) \cdot \frac{1}{x} du = n \cdot \varphi(x)$$

$$\int_0^x \varphi(u) \cdot du = nx \cdot \varphi(x)$$

$$\varphi(x) = n \cdot \varphi(x) + nx \varphi'(x)$$

$$nx \frac{d\varphi}{dx} = (1-n) \varphi$$

$$\frac{d\varphi}{(1-n) \cdot \varphi} = \frac{1}{nx} dx$$



$$10. \quad y' + p(x) \cdot y = q(x) \quad y_1'(x) + p(x) y_1(x) = q(x) \quad \frac{1}{1-n} \cdot \ln|y| = \frac{1}{n} \ln|x| + C$$

$$y' + p(x) \cdot y = 0 \quad y_0'(x) + p(x) \cdot y_0(x) = 0 \quad y^{\frac{1}{1-n}} = C \cdot x^{\frac{1}{n}}$$

$$(y_1(x) + y_0(x))' + p(x)(y_1(x) + y_0(x))$$

$$= y_1'(x) + p(x)y_1(x) + y_0'(x) + p(x)y_0(x)$$

$$= q(x) + 0 = q(x)$$

13. 求解下列微分方程:

(1)  $x^2 y'' = y'^2$ ;

(3)  $y''(e^x + 1) + y' = 0$ ;

(2)  $y'^2 + 2yy'' = 0$ ;

④  $y''' = 2(y'' - 1)\cot x$ .

13. (1)  $p = y'$   
 $y'' = \frac{dp}{dx} \quad x^2 \frac{dp}{dx} = p^2$

$\therefore \frac{dp}{p^2} = \frac{dx}{x^2}$

$\frac{1}{p} = \frac{1}{x} + C_1 = \frac{C_1 x + 1}{x}$

$y' = \frac{x}{C_1 x + 1} \quad dy = \frac{x}{C_1 x + 1} \cdot dx$

$\frac{\frac{1}{C_1}(C_1 x + 1 - 1)}{C_1 x + 1} = \frac{1}{C_1} - \frac{1}{C_1} \frac{1}{C_1 x + 1}$

$\therefore dy = (\frac{1}{C_1} - \frac{1}{C_1} \frac{1}{C_1 x + 1}) dx$

$y = \frac{1}{C_1} x - \ln|C_1 x + 1| \cdot \frac{1}{C_1^2} + C_2$

(2)  $p = y', \quad y'' = \frac{dp}{dx} = \frac{dp}{dy} \cdot \frac{dy}{dx} = p \frac{dp}{dy}$

$p^2 + 2y \cdot p \cdot \frac{dp}{dy} = 0,$

①  $p = 0, \quad y' = 0, \quad y = C$

②  $p \neq 0, \quad p + 2y \frac{dp}{dy} = 0,$

$2y \frac{dp}{dy} = -p, \quad \frac{dp}{p} = \frac{-1}{2y} dy$

$\ln|p| = -\frac{1}{2} \ln|y| + C$

$p = C_1 y^{-\frac{1}{2}} = y'$

$\frac{dy}{dx} = C_1 y^{-\frac{1}{2}}$

$y^{\frac{1}{2}} dy = C_1 dx$

$\frac{2}{3} \cdot y^{\frac{3}{2}} = C_1 x + C_2$

$y^3 = (C_1 x + C_2)^2$

11.

12. 证明: 线性非齐次方程(9.16)的通解  
 $y_1(x) + Ce^{-\int_{x_0}^x p(t)dt}$  (其中  $C$  为任意常数)  
 §3 微分方程解的存在和唯一性定理  
 包含了其一切解.

$y_1(x)$  为 (9.16) 特解

$y_1'(x) + p(x)y_1(x) = q(x)$

设  $y_x(x)$  为一个特解,

$y_x'(x) + p(x)y_x(x) = q(x)$ , 作差,

$(y_1'(x) - y_x'(x)) + p(x)(y_1(x) - y_x(x)) = 0$

取  $y_0(x) = y_1(x) - y_x(x)$ , 则

$y_0'(x) + p(x) \cdot y_0(x) = 0$

两边同乘  $e^{\int_{x_0}^x p(t)dt}$ ,

则  $[y_0(x) \cdot e^{\int_{x_0}^x p(t)dt}]' = 0$

$\therefore [y_1(x) - y_x(x)] \cdot e^{\int_{x_0}^x p(t)dt} = C$

$\therefore y_x(x) = y_1(x) - Ce^{\int_{x_0}^x p(t)dt},$

则  $y_x(x)$  包含在通解中

(3)  $y''(e^x + 1) + y' = 0$

$y' = p,$

考虑  $p = 0$  的情况

$\frac{dp}{dx} \cdot (e^x + 1) = -p.$

$\frac{dp}{-p} = \frac{dx}{e^x + 1} = \frac{e^x}{e^x(e^x + 1)} dx = \frac{1}{e^x(e^x + 1)} de^x \quad \ln|e^x| - \ln|e^x + 1|$

$-\ln|p| = \ln \frac{e^x}{e^x + 1} + C$



$$\therefore p = C_1 \frac{e^x + 1}{e^x} = y'$$

$$\therefore y' = C_1 (1 + e^{-x})$$

$$\therefore y = C_1 [x - e^{-x}] + C_2$$

$$13. (4) \quad y''' = 2(y'' - 1) \cot x \quad \text{令 } y'' = z, \quad y''' = z'$$

$$z' = 2(z - 1) \cot x$$

$$\frac{dz}{dx} = 2(z - 1) \cot x$$

$$\textcircled{1} z \neq 1 \text{ 时, } \frac{dz}{2(z-1)} = \cot x \cdot dx$$

$$\frac{1}{2} \ln|z-1| = \ln|\sin x| + C$$

$$z - 1 = C \sin^2 x \quad \text{则 } y'' = C_1 \sin^2 x + 1$$

$$y' = C_1 \left( \frac{1}{2} x - \frac{1}{4} \sin 2x \right) + x + C_2$$

$$y = C_1 \left( \frac{1}{4} x^2 + \frac{1}{8} \cos 2x \right) + \frac{1}{2} x^2 + C_2 x + C_3$$

②  $z = 1$  时, 亦满足题意,

$$y'' = 1, \quad y' = x + C_1, \quad y = \frac{1}{2} x^2 + C_1 x + C_2$$

可以计入情况①中

$$\text{综上, } y = C_1 \left( \frac{1}{4} x^2 + \frac{1}{8} \cos 2x \right) + \frac{1}{2} x^2 + C_2 x + C_3$$

$$14. (4) \quad (1 + x\sqrt{x^2 + y^2}) dx + (-1 + \sqrt{x^2 + y^2}) y dy = 0$$

$$\frac{\partial Q}{\partial x} = \frac{xy}{\sqrt{x^2 + y^2}} \equiv \frac{\partial P}{\partial y} = \frac{xy}{\sqrt{x^2 + y^2}}, \text{ 是全微分方程}$$

$$\int (1 + x\sqrt{x^2 + y^2}) dx = x + \frac{1}{3} (x^2 + y^2)^{\frac{3}{2}} + C_1$$

$$f(x, y) = x + \frac{1}{3} (x^2 + y^2)^{\frac{3}{2}} + \varphi(y)$$

$$\frac{\partial f}{\partial y} = \frac{1}{2} \cdot 2y \cdot \sqrt{x^2 + y^2} + \varphi'(y) = -y + y \cdot \sqrt{x^2 + y^2}$$

$$\therefore \varphi'(y) = -y, \quad \varphi(y) = -\frac{1}{2} y^2 + C$$

$$\therefore f(x, y) = x + \frac{1}{3} (x^2 + y^2)^{\frac{3}{2}} - \frac{1}{2} y^2 + C$$



$$15. (5) (x dy + y dx) \sqrt{1-y^2} + xy dy = 0 \quad (xy \neq 0, y \neq \pm 1)$$

$$\sqrt{1-y^2} dx + xy dy = 0$$

$$\frac{1}{xy} dx + \frac{1}{\sqrt{1-y^2}} dy = 0$$

$$d \ln |xy| + d \arcsin y = 0$$

$$\therefore \ln |xy| + \arcsin y = C$$

$$16. (1) x dx = (2xy^3 dx + 3x^2 y^2 dy) \sqrt{1+x^2}$$

$$x dx = \sqrt{1+x^2} \cdot d(x^2 y^3)$$

$$\therefore d\sqrt{1+x^2} = d(x^2 y^3)$$

$$\therefore \sqrt{1+x^2} - x^2 y^3 = C.$$

$$16) e^x dx + (e^x \cot y + 2y \cos y) dy = 0$$

$$\frac{\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}}{P} = \frac{e^x \cot y}{e^x} = \cot y$$

$$\therefore e^{\int \cot y dy} = \sin y \text{ 是一个积分因子.}$$

$$e^x \sin y dx + (e^x \cos y + 2y \cos y \sin y) dy = 0$$

$$\int e^x \sin y dx = e^x \sin y.$$

$$\therefore \frac{\partial f}{\partial y} = e^x \cos y + \varphi'(y) = e^x \cos y + 2y \sin y \cos y$$

$$\varphi'(y) = 2y \sin y \cos y = y \sin 2y$$

$$\begin{aligned} \int y \sin 2y dy &= -\frac{1}{2} \int y d(\cos 2y) = -\frac{1}{2} (y \cos 2y - \int \cos 2y dy) = -\frac{1}{2} (y \cos 2y - \frac{1}{2} \sin 2y) \\ &= \frac{1}{4} \sin 2y - \frac{1}{2} y \cos 2y \end{aligned}$$

$$\therefore f(x, y) = e^x \cos y + \frac{1}{4} \sin 2y - \frac{1}{2} y \cos 2y + C$$



14. 判断下列方程是否是全微分方程,若是,求出其通积分.

(1)  $(3x^2+4y)dx+(2x+1)dy=0$ ;

(2)  $(x+2y)dx+(2x-y)dy=0$ ;

(3)  $e^y dx+(xe^y-2y)dy=0$ ;

(4)  $(1+x\sqrt{x^2+y^2})dx+(-1+\sqrt{x^2+y^2})ydy=0$ ;

(5)  $(x+2y)dx+(2x+3y)dy=0$ ;

(6)  $(ax-by)dx+(bx-cy)dy=0$  ( $b\neq 0$ );

(7)  $(ye^x+2e^x+y^2)dx+(e^x+2xy)dy=0$ ;

(8)  $(ax^2+by^2)dx+lxydy=0$  ( $a, b, l$  为常数).

(3)  $xe^y + \varphi(y)$

$xe^y + \varphi'(y) = xe^y - 2y$

$xe^y - y^2$

(5)  $\frac{1}{2}x^2 + 2xy + \frac{3}{2}y^2$

(6)  $x$

(7)  $e^xy + xy^2 + \frac{\varphi(x)}{2e^x}$

(8)  $X$

(3)  $dx + d\ln|\sin y| = 0, d(x + \ln|\sin y|) = 0$

$x + \ln|\sin y| = C$

(4)  $(x^2+y^2)dx + ydx - xdy = 0, (x^2+y^2)dx + (x^2+y^2)d\arctan\frac{y}{x} = 0$

$x + \arctan\frac{y}{x} = C$

16. (2)  $(x^2+y)dx - xdy = 0,$

$\frac{\frac{\partial p}{\partial y} - \frac{\partial q}{\partial x}}{q} = \frac{2}{-x},$  只与x有关

$e^{-2\ln x} = x^{-2}$  是积分因子

$(1+\frac{y}{x^2})dx - \frac{1}{x}dy = 0$

$\int -\frac{1}{x}dy = -\frac{y}{x}, \quad \frac{y}{x^2} + \varphi'(x) = 1 + \frac{y}{x^2}$   
 $\varphi(x) = x + C$

$\therefore f(x,y) = \frac{y}{x^2} + x + C$

14. (1)  $\frac{\partial p}{\partial y} = 4 \neq \frac{\partial q}{\partial x}$

(2)  $\frac{1}{2}x^2 + 2xy + \varphi(y)$

$2x + \varphi'(y) = 2x - y, \varphi(y) = -\frac{1}{2}y^2$

$\therefore f(x,y) = \frac{1}{2}x^2 + 2xy - \frac{1}{2}y^2$

15. (1)  $(x+y)^2: d(x-y) = d(x+y)$

$d(x-y) = \frac{d(x+y)}{(x+y)^2} = -d(x+y)^{-1}$

$\therefore d(x-y + \frac{1}{x+y}) = 0$

$\therefore x-y + \frac{1}{x+y} = C$

(2)  $\frac{\frac{\partial p}{\partial y} - \frac{\partial q}{\partial x}}{q} = 2, \therefore e^{2x}$  是一个积分因子.

$\frac{1}{2}y^2e^{2x} + \varphi(x)$

$e^{2x}(y^2), \quad \varphi'(x) = e^{2x}(x^2+x+1)$

$\frac{1}{2} \int (x^2+x+1)de^{2x} = \frac{1}{2}e^{2x}(x^2+x+1)$

$-\frac{1}{2} \int e^{2x}d(x^2+x+1)$

$= \frac{1}{2}[e^{2x}(x^2+x+1) - (2x+1)e^{2x}]dx$

$= \frac{1}{2}[e^{2x}(x^2+x+1) - \frac{1}{2}(2x+1)e^{2x}]dx$

$= \frac{1}{2}[e^{2x}(x^2+x+1) - \frac{1}{2}(2x+1)e^{2x}]dx$

$+ \int 10e^{2x}dx$

$= \frac{1}{2} \cdot (e^{2x}(x^2+x+1) - \frac{1}{2}(2x+1)e^{2x}) + \frac{1}{2}e^{2x}$



16. 利用积分因子, 求解下列微分方程:

①  $x dx = (2xy^3 dx + 3x^2 y^2 dy) \sqrt{1+x^2}$ ;

(2)  $(x^2 + y) dx - x dy = 0$ ;

(3)  $y(x+1) dx + x(y+1) dy = 0$  ( $xy \neq 0$ );

(4)  $(3x^2 y + 2xy + y^3) dx + (x^2 + y^2) dy = 0$ ;

(5)  $2xy^3 dx + (x^2 y^2 - 1) dy = 0$ ;

⑥  $e^x dx + (e^x \cot y + 2y \cos y) dy = 0$ .

(3)  $xy(dx+dy) + ydx + xdy = 0$

$$xy d(x+y) + dxy = 0$$

$$d(x+y) + \frac{1}{xy} dxy = d(x+y) + d \ln(xy) = 0$$

$$\therefore x+y + \ln xy = C$$

(4)  $\frac{\partial P}{\partial y} = 3x^2 + 2x + 3y^2$

$$\frac{\partial Q}{\partial x} = 2x$$

$$\frac{\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}}{Q} = 3, \therefore e^{3x} \text{ 是积分因子.}$$

$$e^{3x} (3x^2 y + 2xy + y^3) dx + (x^2 + y^2) e^{3x} dy = 0$$

$$(x^2 y + \frac{1}{3} y^3) \cdot e^{3x} + \varphi(x)$$

$$2xy e^{3x} + (3x^2 y + y^3) \cdot e^{3x}$$

$$\circ (x^2 y + \frac{1}{3} y^3) \cdot e^{3x} = C$$

(5)  $\frac{\partial P}{\partial y} = 6xy^2, \frac{\partial Q}{\partial x} = 2xy^2$

$$\frac{\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}}{P} = \frac{-4xy^2}{2xy^3} = -2y^{-1}, \text{ 只与 } y \text{ 有关}$$

$$\text{则 } e^{-2 \ln |y|} = y^{-2} \text{ 是积分因子}$$

$$2xy dx + (x^2 - \frac{1}{y^2}) dy = 0$$

$$x^2 y + \varphi(y) \quad x^2 + \varphi'(y) = x^2 - \frac{1}{y^2}$$

$$x^2 y + \frac{1}{y} = C$$



# 习 题 9.3

1. 设函数  $f(x, y)$  关于  $y$  的偏导数  $f_y(x, y)$  在闭矩形域  $R$  上有界, 证明:  $f(x, y)$  在  $R$  上对  $y$  满足利普希茨条件.
2. 下列函数在相应的闭区域上是否满足利普希茨条件?
  - (1)  $f(x, y) = \sqrt{y}$ ,  $R: |x| \leq a, 0 \leq y \leq b$ ;
  - (2)  $f(x, y) = \sqrt{y}$ ,  $R: |x| \leq a, 0 < \delta \leq y \leq b$ ;
  - (3)  $f(x, y) = xy^n$  ( $n$  为正整数),  $R: |x| \leq a, |y| \leq b$ ;
  - (4)  $f(x, y) = xy^2$ ,  $D: |x| \leq a, -\infty < y < +\infty$ .
3. 我们将本节中的存在和唯一性定理应用于下列情况:  $f(x, y) = x^2 y + x, x_0 = 0, y_0 = 0, R = \{(x, y) \mid |x| \leq 1, |y| \leq 1\}$ . 试估计利普希茨常数  $L$  及  $M = \max_R |f(x, y)|$  的值, 并给出定理中解存在的区间长度  $2h$  的估计值.
4. 求初值问题

$$\begin{cases} \frac{dy}{dx} = x + y, \\ y(0) = -2 \end{cases}$$

的皮卡序列及其极限.

5. 求初值问题

$$\begin{cases} \frac{dy}{dx} = x - y^2, \\ y(0) = 0 \end{cases}$$

的皮卡序列的近似解:  $y_0(x), y_1(x)$  及  $y_2(x)$ .

6. 根据解的存在和唯一性定理, 指出下列各初值问题的解的存在区间:

$$(1) \begin{cases} y' = \sqrt{x-y}, \\ y(4) = 1, \end{cases} \quad R: |x-4| \leq 2, |y-1| \leq 1;$$

$$(2) \begin{cases} y' = \frac{\sqrt{y-x}}{x-2}, \\ y(0) = 3, \end{cases} \quad R: |x| \leq 1, |y-3| \leq 1;$$

$$(3) \begin{cases} \frac{dy}{dx} = x + e^x, \\ y(1) = 0, \end{cases} \quad R: |x-1| \leq 1, |y| \leq 1.$$

$$4. \quad y = -2 + \int_{x_0}^x (x+y(x)) dx$$

$$y_1(x) \equiv -2, \quad y = -2 + \int_{x_0}^x (x-2) dx$$

1. 证明下列函数组线性无关:

$$(1) e^{\lambda x}, x e^{\lambda x}; \quad (2) \cos \beta x, \sin \beta x; \quad (3) e^{\alpha x} \cos \beta x, e^{\alpha x} \sin \beta x.$$

2. 设  $\varphi_1(x), \varphi_2(x)$  是微分方程

$$y'' + q(x)y = 0$$

的任意两个解, 其中  $q(x)$  在  $(a, b)$  上连续. 证明:  $\varphi_1(x)$  与  $\varphi_2(x)$  的朗斯基行列式

$$W(x) \equiv \text{常数}, \quad x \in (a, b).$$

3. 设  $y = \varphi(x)$  是线性齐次微分方程 (9.61) 的一个 非零解 (即  $\varphi(x) \neq 0, x \in (a, b)$ ), 若有  $x_0 \in (a, b)$  使  $\varphi(x_0) = 0$ , 证明:  $\varphi'(x_0) \neq 0$ .

4. 设  $\varphi_1(x)$  与  $\varphi_2(x)$  是线性齐次微分方程 (9.61) 的两个线性无关的解. 证明: 它们没有公共的零点.

5. 证明: 线性齐次微分方程 (9.61) 存在两个线性无关的解  $y = \varphi_1(x)$  及  $y = \varphi_2(x)$ .

6. 证明: 线性非齐次微分方程 (9.59) 的任一解, 都包含在它的通解 (9.69) 式中.

..

$$3. \quad \text{设 } \varphi_1(x), \varphi_2(x) \text{ 为 9.61 的两个解}$$

$$\text{则必有 } \varphi(x) = C_1 \varphi_1(x) + C_2 \varphi_2(x)$$

$$\text{且 } W(x) = \begin{vmatrix} \varphi_1(x) & \varphi_2(x) \\ \varphi_1'(x) & \varphi_2'(x) \end{vmatrix} = \varphi_1(x) \varphi_2'(x) - \varphi_2(x) \varphi_1'(x) \neq 0$$

$$\text{而若 } \varphi(x_0) = 0 \text{ 且 } \varphi'(x_0) = 0$$

$$9.3. \quad 1. \quad f(x_1, y_1 + \Delta y) = \frac{\partial f}{\partial y} \cdot (y_1 + \theta \Delta y) \cdot \Delta y + f(x_1, y_1)$$

$$\text{令 } \Delta y = y_2 - y_1, \text{ 则}$$

$$f(x_1, y_2) = \frac{\partial f}{\partial y} \cdot (y_1 + \theta \Delta y) \cdot (y_2 - y_1) + f(x_1, y_1)$$

$$\therefore |f(x, y_2) - f(x, y_1)| = \left| \frac{\partial f}{\partial y} (y_1 + \theta \Delta y) \right| \cdot |y_2 - y_1|$$

$$2. \quad (2)$$

$$f(x, y_2) - f(x, y_1) = x(y_2^n - y_1^n) = x \cdot (y_2 - y_1)(y_2^{n-1} + y_2^{n-2}y_1 + \dots + y_1^{n-1})$$

$$\therefore |f(x, y_2) - f(x, y_1)| = |y_2 - y_1| \cdot |x(y_2^{n-1} + \dots + y_1^{n-1})| \leq |y_2 - y_1| \cdot a(n+1) \cdot b^{n-1}.$$

$$6. \quad y(x) = C_1 \varphi_1(x) + C_2 \varphi_2(x) + \underline{y^*(x)}$$

$$y'' + p(x)y' + q(x)y = f(x)$$

$$\text{设 } y_0(x) \text{ 是一个解, 即}$$

$$y_0''(x) + p(x)y_0' + q(x)y_0 = f(x)$$



即 
$$\begin{cases} \varphi_1(x) C_1 + \varphi_2(x) C_2 = 0 \\ \varphi_1'(x) C_1 + \varphi_2'(x) C_2 = 0 \end{cases}$$

必有解

由线代知识, 有  $\begin{vmatrix} \varphi_1(x) & \varphi_2(x) \\ \varphi_1'(x) & \varphi_2'(x) \end{vmatrix} = 0$ , 与已知矛盾